Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/29/2013
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Expedition 19/20 NOTE: End date changed to 5/31/2013, per NSSC (Ed., 6/6/12) NOTE: End date is now 5/31/2012 (from 5/31/2010) per NSSC information and F. Szofran/MSFC (10/2009) NOTE: Changed to FLIGHT project, per F. Szofran/MSFC (10/2009) NOTE: End date changed to 5/31/2010 (from 5/31/2009) per PI (12/08)		
Key Personnel Changes/Previous PI:	August 2013 report: Nick Mauro finished his PhD and moved to a research project funded by my NSF-MRI, designing and constructing a new electrostatic levitation facility that will be installed on the Spallation Neutron Source at Oak Ridge National Laboratory. Two new students, Matt Blodgett and Jennifer Gewin, were moved onto this NASA project. Dr. Anup Gangopadhyay continued to be partially supported by this research grant.		
COI Name (Institution):	Herlach, Dieter (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation) Hyers, Robert (University of Massachusetts)		
Grant/Contract No.:	NNX07AK27G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Supercooling, the ability of a liquid held at temperatures below its freezing temperature without crystallizing, was first discovered by Fahrenheit in 1724 when he was studying the freezing of water. However, it wasn't until 1952 that supercooling in liquid metals was demonstrated by David Turnbull. That liquids can be supercooled indicates that there is a barrier to the crystallization of the liquid; this is called the nucleation barrier. Nucleation is the first step in most first-order phase transitions, like the freezing of water. Nucleation is important in processes ranging from the crystallization of liquids to the development of bubbles in carbonated beverages to biological processes. It is even believed to be central in phase transitions that occurred in the early universe. An ability to control nucleation is required for the development of new materials for advanced applications. In almost all cases nucleation data are described by the classical theory of nucleation, first developed in the late 19th century. It is now recognized however that there are serious problems with aspects of that theory in some regimes of application. Also, it does not properly describe nucleation when the chemical concentrations of the initial and final phases are different. Finally, recent evidence indicates that nucleation is often coupled to other processes and phase transitions, something that cannot be described readily within the classical theory. Understanding nucleation requires understanding not only the complicated nucleation kinetic processes, but also the relevant thermodynamic parameters and the relation between structure and these properties. The task of the continuing ground-based research on this contract is the investigation of a coupling between structural ordering in the liquid and associated thermophysical properties, and with the nucleation barrier, as well as with other phase transitions that are being discovered that occur in supercooled liquids. Studies are focused on transition metals and their alloys. Ground-based measurements of the structure, specific heat, density, viscosity, surface tension, and emissivity in equilibrium and supercooled liquids are made using electrostatic levitation containerless processing techniques. In cooperation with ESA these data will be combined with studies made on the International Space Station to develop a better understanding of the complicated nucleation processes, and other phase transitions, that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The WU-BESL (Washington University Beamline Electrostatic Levitation) technique, developed under partial support by this contract, allows a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions. The discovery of possible new phase transitions in the liquid will lead to a better understanding of glass formation. The modification of WU-BESL to also include density, viscosity surface tension and specific heat measurements, in addition to structural data from synchrotron X-ray diffraction measurements, is allowing novel investigations of supercooled metallic liquids that are leading to a better understanding of how liquids order and how this relates to glass formation.
	In 2005, NASA priorities shifted regarding microgravity experiments on the International Space Station, causing many planned flight experiments to be put on hold, including this one - QUASI. NNX07AK27G was awarded for continued ground-based research and the potential of future flight opportunities. Those opportunities did materialize, as part of a co-operative study with the THERMOLAB team, which is scheduled to carry out ESA-funded studies on the ISS. QUASI was invited to collaborate on samples that had already been selected by ESA and was also given a new sample that was directly relevant to the questions that QUASI had originally planned to address. Currently, our ISS experiments are planned for 2015. The scope will be to investigate the effects of stirring and microgravity on the nucleation and growth of crystal phases, testing a new model that we have developed for diffusion-limited nucleation. Thermophysical properties will also be measured on the ISS, particularly the specific heat, for which more quantitative data can be obtained than on earth. The property measurements will complement ground-based structural studies of the supercooled liquids, allowing presumed connections between structure, properties and phase transitions to be explored. They will also provide parameters that are needed for a quantitative analysis of the data obtained from the nucleation measurements. This is the last year of this grant. Over the entire course of the grant, activities have included: • The construction and improvement of an ElectroStatic Levitation (ESL) facility at Washington University, the Washington University Beamline ESL (WU-BESL). • Structural studies of over one hundred different equilibrium and supercooled high temperature liquid metals in a containerless environment. The scattering studies were made at the Advanced Photon Source using WU-BESL. • Development of more quantitative analysis methods for the X-ray scattering data, which include the proper sample geometry. • Measurements of thermophysical properties of these metallic liquids, including the density, surface tension, viscosity and specific heat and correlation of these properties with structural evolution, phase transitions and other fundamental aspects of the liquid, in particular liquid fragility. • Demonstration that WU-BESL can be used to determine phase diagrams for high temperature alloys, such as hafnium (Hf) and rhodium (Rh) and hafnium and platinum (Pt) near the eutectic composition (where the liquidus temperature has a local, and possibly global, minimum, depending on the details of the phase diagram). In many cases these phase diagrams are poorly known or do not even exist. • Supporting regular Internal Working Group (IWG) meetings with the other members of the ISS team. • Measuring the evaporation rates of the liquids in WU-BESL for our samples and for some samples from other members of the IWG. These are necessary for calculations of the expected evaporation on the coil of the Electromagnetic Levitation facility (EML) that will be used on the ISS, and for estimates of the dust generated, which if released into the ISS cabin would pose a health risk for the astronauts. In addition to providing useful information that will be used to interpret the results of the ISS studies, these activities have led to a number of conclusions of fundamental scientific importance: • The structures of transition metal and transition-noble metal alloy liquids have a significant degree of topological

Task Progress:	short-range order (i.e. nearest neighbor), with a local symmetry similar to that of an icosahedron. • Chemical ordering is likely linked to this topological order. • In some cases, there is evidence that the ordering occurs abruptly, possibly signaling a novel phase transition within the liquid. • Liquids containing atoms that have a strong attraction (negative heat of mixing), like the transition-noble metal liquids, develop more extended order, reaching beyond the nearest-neighbor shell (medium-range order). • The rate of structural ordering in the liquid is related to a fundamental concept, liquid fragility, which is itself related to glass formability. Strong liquids, which order continuously from high temperature, tend to form glasses more easily. Fragile liquids, which show little ordering until the temperature approaches the glass transition temperature, tend to be poorer glass formers. These activities and results evolved over the entire course of this grant. Specific activities for this year include the following. 1. Improvements of WU-BESL - Over the past year, a number of improvements have been made to the ESL located at Washington University, including the addition of a more intense ultraviolet lamp for better sample processing, use of high-intensity LEDs instead of He-Ne lasers to improve sample stability, implementation of optics and associated electronics that allow measurements of the liquid viscosity and the development of techniques for modulated calorimetry. WU-BESL is optimized for structural studies at the Advanced Photon Source, designed for easy transportation and having its own stepping-motor controlled table that allows it to be used in a variety of the hutches at the APS. Unfortunately, while the ability to make modulated calorimetry measurements was demonstrated, the precision of the data obtained are not as high as for the more standard free-cooling studies, in which the heating laser power is removed and the sample is allowed to free cool by radiation loss. While reasonable estimates of the specific heat as a function of temperature can be obtained from these data, more quantitative measurements will require use of the EML facility on the ISS. 2. Improvement of the X-ray Analysis Software – A key contribution from our group is to support ground-based X-ray diffraction studies for our samples and samples from other members of the IWG, providing complementary data for the upcoming ISS experiments. While widely used software for the analysis of synchrotron scattering data exists (such as PDFgetX2), these programs are ill suited to the analysis of the liquid scattering data obtained with WU-BESL since they do not incorporate the spherical geometry of our samples. This is due primarily to the lack of a theoretical expression for the absorption and multiple scattering for spherical samples when the beam is smaller than the sample size, which is the case for our measurements. Over the past year we have completed a theoretical analysis of this problem, developing expressions that take this into account and incorporating them into a new analysis software package. 3. Investigation of Cu-Zr liquids - Two Cu-Zr alloys (Cu50Zr50 and Cu46Zr54) have been selected for study on the ISS in the Batch 2 samples. Our ground-based work has shown evidence for the rapid onset of chemical ordering and an associated specific heat maximum at temperatures far above the glass transition temperature, which could indicate the presence of a liquid/liquid phase transition. Such a transition is believed by some to occur in supercooled metallic liquids, but it has never been confirmed. To investigate this further, we have made synchrotron studies on a range of Cu-Zr liquids of different composition. From these we have constructed the total pair correlation function, g(r), which is a measure of the local topological order. The total g(r) is constructed from three sets of partial pair correlations functions, which for Cu-Zr describe the structural relations between Cu-Cu atoms (g(r)Cu-Cu), Cu-Zr atoms (g(r)Cu-Zr) and Zr-Zr atoms (g(r)Zr-Zr). Interestingly, the total g(r)s for all compositions studied can be fit with the same set of partials. This indicates that changes observed in g(r) are only due to the weighting factors used when the partials are summed to give the total pair correlation function. This can have a significant importance for understanding the chemical ordering across the compositional series, but it also means that if molecular dynamics models exist for one composition of an alloy system, they can be used (when properly weighted) to model the behavior at other compositions. 4. Studies of Vit 106 and Vit 106a - The Vit 106 series of alloys are known for their ability to easily form metallic glasses. The Vit 106 and Vit 106a alloys are planned for measurements in Batch 1 of the ISS studies. To support those studies, we have measured thermophysical properties using WU-BESL, including the specific volume, thermal expansion coefficient and specific heat. The specific heat and thermal expansion as a function of temperature support predications from an energy landscape argument that we made last year describing the relation between liquid fragility and thermal expansivity at high temperatures, far above the glass transition temperature and near the melting temperature. However, they also cast doubt on a recent report by others of a liquid/liquid phase transition in the Vit 106. Our studies indicate that this is actually the nucleation and slow growth of a crystal phase within the liquid. The work reported this year for this grant and that from previous years is laying the foundation for a better understanding of the links between structure, physical properties and phase transitions in supercooled liquids. The results have both basic and practical significance. These investigations will be taken to the next step in planned ISS experiments that will be supported by the successor to this grant, NNX10AU19G.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Mauro NA, Fu W, Bendert JC, Cheng YQ, Ma E, Kelton KF. "Local atomic structures in equilibrium and supercooled liquid Zr75.5Pd24.5." Journal of Chemical Physics. 2012 Jul;137(4):044501. http://dx.doi.org/10.1063/1.4737381 , Jul-2012
Articles in Peer-reviewed Journals	Mauro NA, Johnson ML, Bendert JC, Kelton KF. "Structural evolution in Ni-Nb and Ni-Nb-Ta liquids and glasses--A measure of liquid fragility?" Journal of Non-Crystalline Solids. 2013 Feb 15;362:237-45. http://dx.doi.org/10.1016/j.jnoncrysol.2012.11.022 , Feb-2013
Articles in Peer-reviewed Journals	Mauro NA, Kelton KF. "Medium range order in Zr-noble metal eutectic liquids." Journal of Non-Crystalline Solids. 2012 Dec 1;358(23):3057-9. http://dx.doi.org/10.1016/j.jnoncrysol.2012.08.006 , Dec-2012
Articles in Peer-reviewed Journals	Bendert JC, Gangopadhyay AK, Mauro NA, Kelton KF. "Volume expansion measurements in metallic liquids and their relation to fragility and glass forming ability: an energy landscape interpretation." Physical Review Letters. 2012 Nov 2;109(18):185901. http://dx.doi.org/10.1103/PhysRevLett.109.185901 , Nov-2012

Articles in Peer-reviewed Journals	Gangopadhyay AK, Bendert JC, Mauro NA, Kelton KF. "Inverse correlation between cohesive energy and thermal expansion coefficient in liquid transition metal alloys." Journal of Physics: Condensed Matter. 2012 Sep;24(37):375102. http://dx.doi.org/10.1088/0953-8984/24/37/375102 , Sep-2012
Articles in Peer-reviewed Journals	Gangopadhyay AK, Hyers RW, Kelton KF. "Nucleation and thermophysical properties of glass-forming liquids." Journal of Metals. 2012 Sep;64(9):1109-17. http://dx.doi.org/10.1007/s11837-012-0422-1 , Sep-2012
Articles in Peer-reviewed Journals	Kelton KF. "Crystal nucleation in supercooled liquid metals." International Journal of Microgravity Science and Applications. 2013;30(1):11-8. , Jan-2013
Articles in Peer-reviewed Journals	Bendert JC, Blodgett ME, Kelton KF. "Calculation of absorption and secondary scattering of X-rays by spherical amorphous materials in an asymmetric transmission geometry." Acta Crystallographica Section A. 2013 Mar;69(2):131-9. http://dx.doi.org/10.1107/S0108767312046272 , Mar-2013
Articles in Peer-reviewed Journals	Bendert JC, Kelton KF. "Correlation between kinetic strength, volumetric properties, and glass forming ability in metallic liquids." Journal of Non-Crystalline Solids. 2013 Sep 15;376:205–8. http://dx.doi.org/10.1016/j.jnoncrysol.2013.05.032 , Sep-2013
Articles in Peer-reviewed Journals	Bendert JC, Blodgett ME, Gangopadhyay AK, Kelton KF. "Measurement of volume, thermal expansion and specific heat in Zr57Cu15.4Ni12.6Al10Nb5 and Zr58.5Cu15.6Ni12.8Al10.3Nb2.8 liquids and glasses." Applied Physics Letters. 2013 May;102(21):211913. http://dx.doi.org/10.1063/1.4808030 , May-2013
Articles in Peer-reviewed Journals	Soklaski R, Nussinov Z, Markow Z, Kelton KF, Yang L. "Connectivity of the icosahedral network and a dramatically growing static length scale in Cu-Zr binary metallic glasses." Physical Review B. 2013 May;87(18):184203. http://dx.doi.org/10.1103/PhysRevB.87.184203 , May-2013
Articles in Peer-reviewed Journals	Mauro NA, Vogt AJ, Johnson ML, Bendert JC, Kelton KF. "Anomalous structural evolution in Cu50Zr50 glass-forming liquids." Applied Physics Letters. 2013 Jul;103(2):021904. http://dx.doi.org/10.1063/1.4813389 , Jul-2013
Articles in Peer-reviewed Journals	Bendert JC, Mauro NA, Kelton KF. "Pair distribution function analysis of X-ray diffraction from amorphous spheres in an asymmetric transmission geometry: application to a Zr58.5Cu15.6Ni12.8Al10.3Nb2.8 glass." Journal of Applied Crystallography. 2013 Aug;46(4):999-1007. http://dx.doi.org/10.1107/S0021889813013162 , Aug-2013
Articles in Peer-reviewed Journals	Bendert JC, Pueblo CE, Veligati S, Mauro NA, Kelton KF. "Temperature Calibration for Optical Pyrometry in Containerless Systems using Differential Scanning Calorimetry - Application to Cu100-xZrx (x=45-50)." International Journal of Thermophysics. (in press, as of August 2013). , Aug-2013
Articles in Peer-reviewed Journals	Bendert JC, Kelton KF. "Containerless measurements of density and viscosity for a Cu48Zr52 liquid." International Journal of Thermophysics. (in press, as of August 2013). , Aug-2013
Articles in Peer-reviewed Journals	Bradshaw RC, Hyers RW, Rogers JR, Rathz TJ, Lee GW, Gangopadhyay AK, Kelton KF. "Non-contact techniques for measurement of density: data for a quasicrystal-forming Ti-Zr-Ni alloy." International Journal of Thermophysics. (in press, as of August 2013). , Aug-2013
Articles in Peer-reviewed Journals	Blodgett M, Gangopadhyay AK, Kelton KF. "Thermal evaporation measurements for quasicrystal (TIZrNi) and glass-forming (Vit 106 and Vit 106a) liquids." International Journal of Thermophysics. (in press, as of August 2013). , Aug-2013
Dissertations and Theses	Bendert JC. "Thermophysical and Structural Measurements of Liquid Metallic Alloys Using Electrostatic Levitation." PhD Dissertation, Washington University, St. Louis, MO, May 2013. , May-2013
NASA Technical Documents	Kelton KF, Bendert JC, Mauro NA. "Liquid structures and physical properties - ground based studies for ISS experiments." In: Materials Research in Microgravity 2012. R. Hyers, et al., Eds. Proceedings of a symposium held at the 141st TMS 2012 Annual Meeting and Exhibition, Orlando, Florida, March 11–15, 2012. Huntsville, AL : NASA Marshall Space Flight Center, 2012. NASA/CP—2012–217466. p. 33-40. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20120015582_2012018023.pdf , Sep-2012